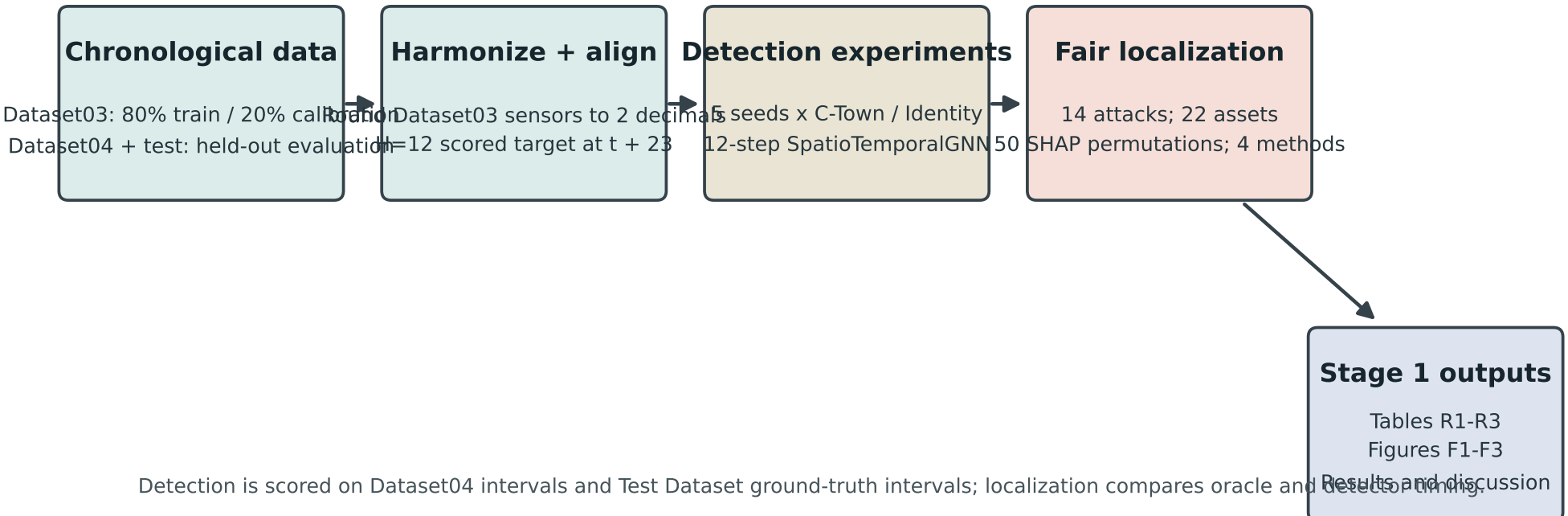Stage 1 experimental pipeline